REPORT

VIT BHOPAL UNIVERSITY

Course Code: CSA2001
Course Name : Fundamentals in AI and ML

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Branch : CSE - AIML
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Project Report: Emotion-Based Song Recommendation System

1. Project Overview

- **Project Name:** Emotion-Based Song Recommendation System
- **Technology Stack:** Python, Natural Language Processing (NLTK), Machine Learning (Scikit-learn)
- **Type:** Artificial Intelligence / Machine Learning Application
- **Author:** Ananya Sinha (25BAI11375)
- **Course:** Artificial Intelligence and Machine Learning
- **Institution:** VIT Bhopal University

Description:

The Emotion-Based Song Recommendation System is an intelligent AI application designed to detect human emotions from textual input and recommend songs based on the identified emotional state. The system applies Natural Language Processing (NLP) techniques to preprocess textual data and utilizes a Machine Learning classification algorithm to categorize emotions.

After predicting the emotional category such as joy, sadness, anger, love, fear, or surprise, the system recommends songs from a predefined dataset aligned with the detected mood. The application demonstrates the practical use of Artificial Intelligence concepts in building personalized recommendation systems.

Key Functionalities:

• Emotion detection from text input
• NLP preprocessing including text cleaning and stopword removal
• Machine Learning classification model for emotion prediction
• Song recommendation based on detected emotional state
• Structured dataset for mapping emotions to songs
• Interactive command-line interface
• Modular project architecture

2. Project Objectives

The main objectives of this project are:

-To know how Natural Language Processing techniques work.

-To implement text processing techniques like tokenization and removal of stopwords.

-To implement a Machine Learning model for emotion classification.

-To develop a recommendation system based on emotions.

-To show how Artificial Intelligence is used in real-time applications.

-To develop a well-structured AIML project that can be used for academic purposes.

3. Project Development Process

The development process of the Emotion-Based Song Recommendation System was as follows:

a.)Requirement Analysis: Understanding the influence of emotional context on music preference and the ability of the system to identify emotions in text.

b.)Dataset Preparation: Preparing a dataset that contains text samples that are identified by emotions such as joy, sadness, anger, love, fear, and surprise.

c.)NLP Preprocessing: Applying Natural Language Processing techniques such as:

• Converting text to lowercase letters.

• Removing punctuation.

• Removing stopwords.

• Converting text into a machine-readable format.

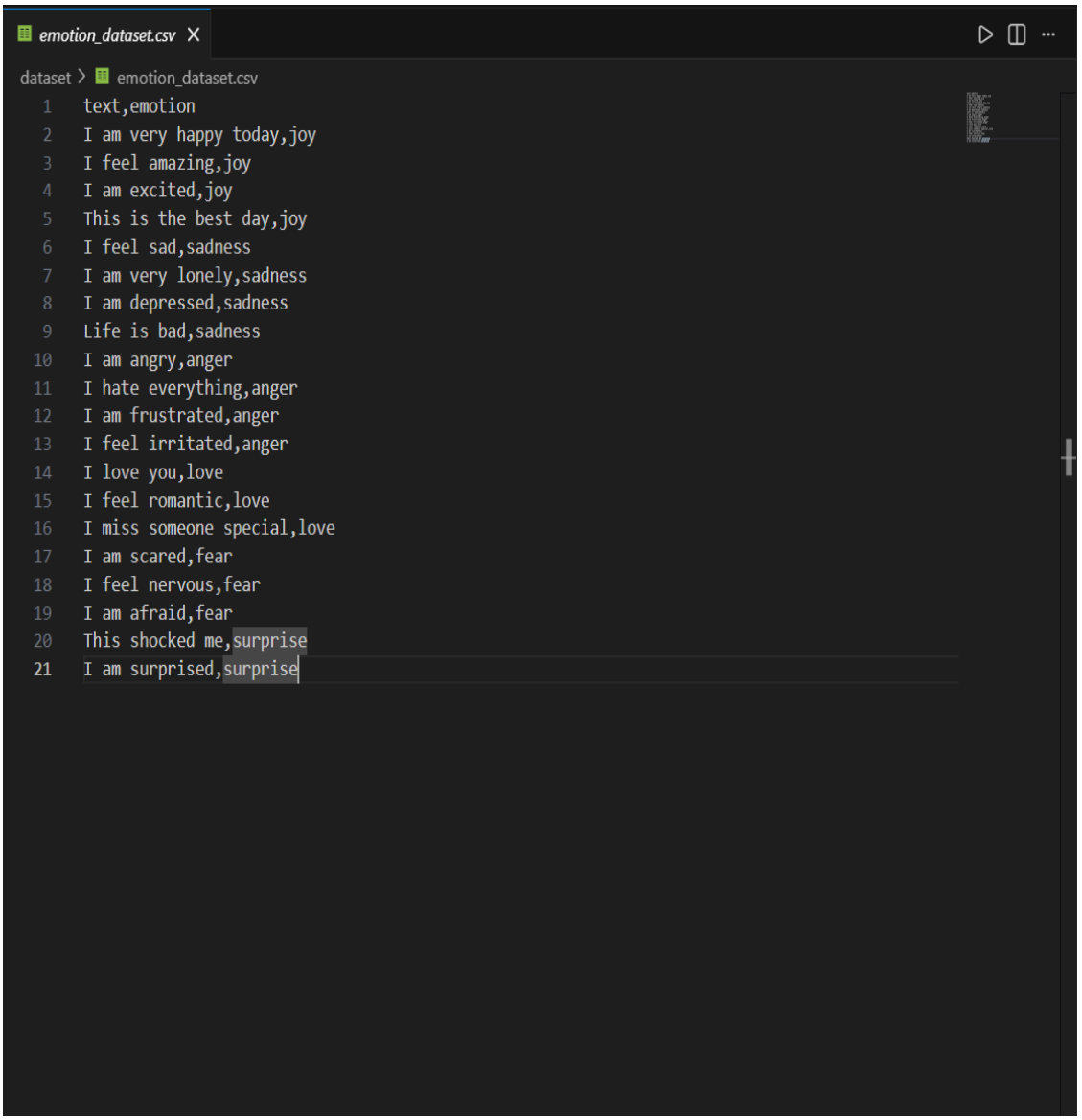
d.)Model Development: Using the TF-IDF Vectorization technique to convert the text into a numerical format and a Machine Learning Classification technique (Naive Bayes) to identify emotions.

e.)Recommendation Logic: Preparing a mapping dataset that identifies the emotion with a set of suitable songs.

f.) Testing and Validation: Testing the system by using different input statements to determine the accuracy of the emotion identification and the relevance of the recommended songs.

g.)Documentation: Preparing a well-structured documentation and a GitHub repository for submission and evaluation.

-Screenshot 1: emotion_dataset.csv open in VS Code



- Screenshot 2: songs_by_emotion.csv open

songs_by_emotion.csv

songs > songs_by_emotion.csv

```csv
emotion,song,artist
joy,Happy,Pharrell Williams
joy,On Top of the World,Imagine Dragons
joy,Good Life,OneRepublic
sadness,Someone Like You,Adele
sadness,Let Her Go,Passenger
sadness,Fix You,Coldplay
anger,Stronger,Kanye West
anger,In The End,Linkin Park
anger,Believer,Imagine Dragons
love,Perfect,Ed Sheeran
love,Tum Mile,Neeraj Shridhar
love,Raabta,Arijit Singh
fear,Demons,Imagine Dragons
fear,Boulevard of Broken Dreams,Green Day
surprise,Counting Stars,OneRepublic
surprise,Shake It Off,Taylor Swift
```

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Purpose: shows dataset creation step

4. System Architecture

Overall Architecture:

User Input → NLP Processing → Machine Learning Model → Emotion Prediction → Song Recommendation Output

a.) NLP Processing Module

Responsibilities:

• converting input text into lowercase format
• removing punctuation marks

• removing stop words using NLTK
• preparing cleaned text for feature extraction

b.) Machine Learning Model

Responsibilities:

• converting text into numerical representation using TF-IDF vectorization
• training classification algorithm on labeled emotion dataset
• predicting emotional category for new text input

Algorithm Used:
Multinomial Naive Bayes

c.) Recommendation Module

Responsibilities:

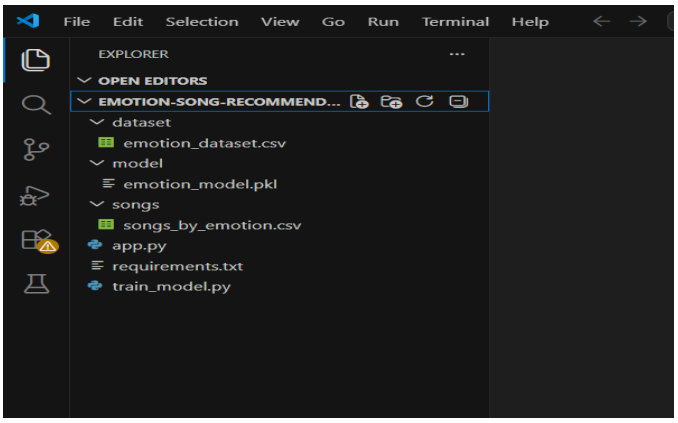
• mapping predicted emotion to song dataset
• selecting a suitable song associated with detected emotion
• displaying song name and artist to the user

d.) Dataset Components

Emotion Dataset:
Contains labeled text samples mapped to emotion categories.

Song Dataset:
Contains emotion-wise categorized song list including artist names.

- Screenshot: VS Code project folder structure



Purpose: shows project architecture implementation

5. Environment and Dependencies

Required libraries are listed in requirements.txt:

pandas
numpy
scikit-learn
nltk
Job lib

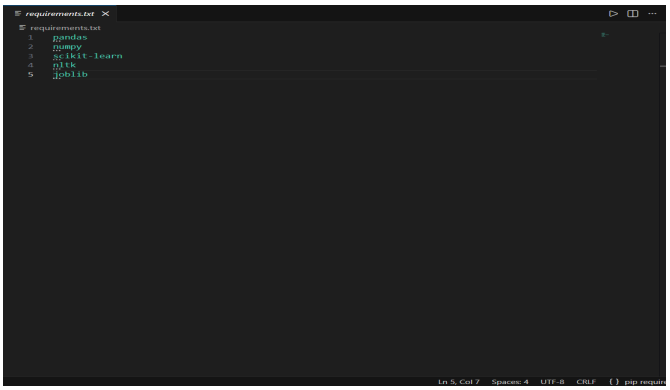
- These libraries support:

• data processing
• machine learning model development
• text preprocessing
• model saving and loading

- Development Environment:
Visual Studio Code

- Programming Language:
Python 3

- Screenshot: requirements.txt open in VS Code



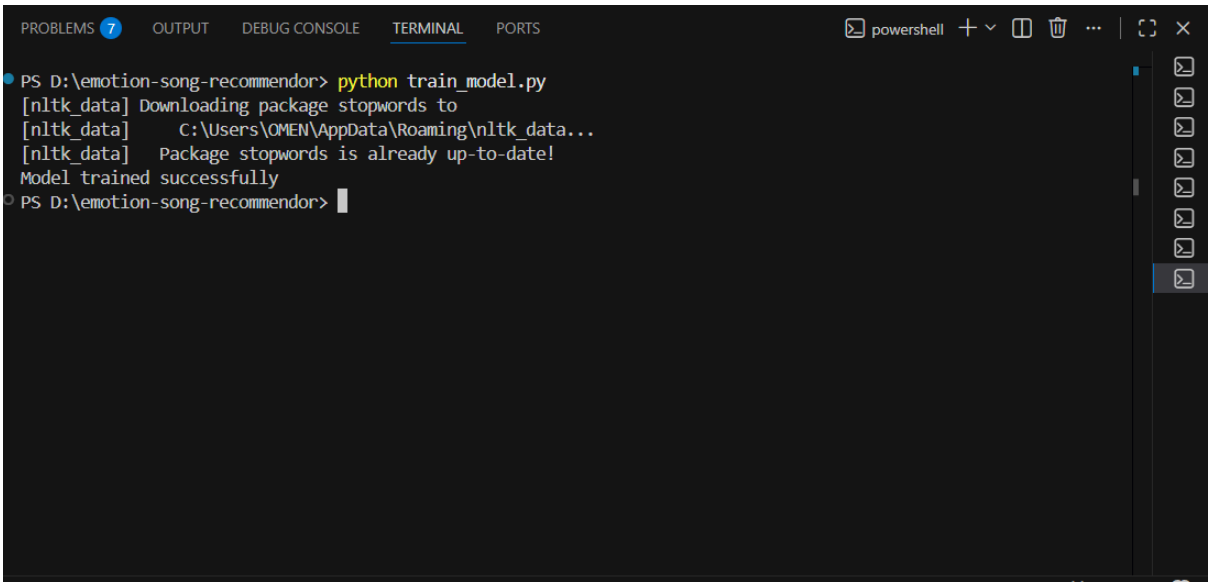
Purpose: shows dependencies used

6. Features

a.) Emotion Detection - Identifies emotion from text
b.) NLP Processing - Cleans and processes textual data
c.) Machine Learning Model - Predicts emotion category

d.) Song Recommendation - Suggests songs based on mood
e.) Dataset Mapping - Emotion to Song
f.) Interactive CLI - User friendly interaction
g.) Modular Structure - Easy to understand project design
h.) Extendable - Can integrate GUI and APPs

- Screenshot: Terminal output showing



Purpose: shows ML model training success

7. How to Run the Application

Step-by-step execution:

1. Open project folder in Visual Studio Code
2. Install dependencies:

pip install -r requirements.txt

3. Train the model:

python train_model.py

4. Run application:

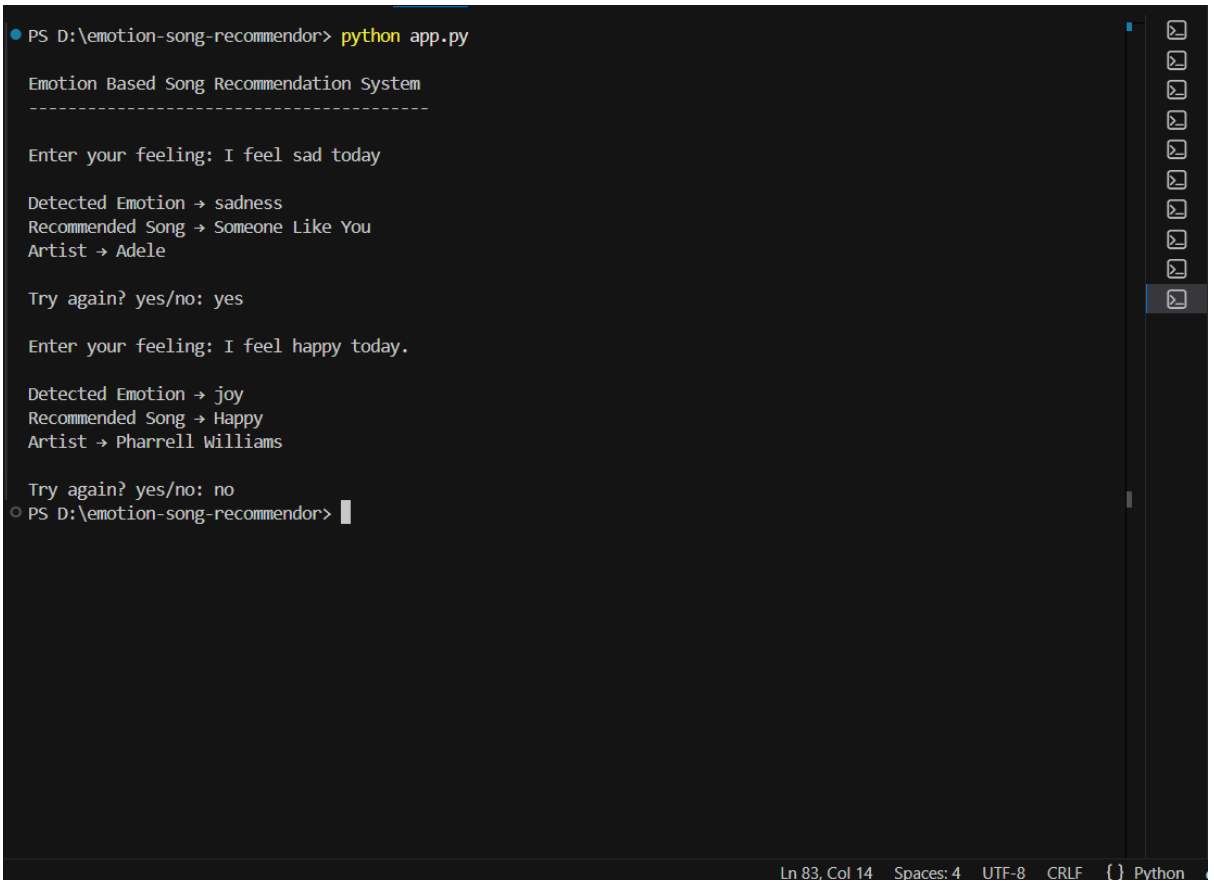
python app.py

5. Enter text describing emotion
6. System will predict emotion and recommend song

8. User Interaction Flow

1. User enters text describing feelings.
2. System preprocesses text using NLP.
3. Machine Learning model predicts emotion.
4. Emotion is matched with song dataset.
5. Recommended song is displayed.
6. User can repeat process for multiple inputs

- Screenshot: Terminal output showing final result.



Purpose: proves project working

9. Project Design and Implementation

Design Considerations:

• simple and readable code structure
• modular implementation of NLP and ML components
• clear separation of dataset, model, and application logic
• efficient data handling using pandas
• scalable architecture for future enhancements

The project is designed in such a way that additional emotions or songs can easily be added to datasets.

10. Conclusion

The Emotion-Based Song Recommendation System project has shown how Artificial Intelligence concepts can be applied in real life in terms of understanding human emotions by analyzing texts.

The project has successfully used Natural Language Processing and Machine Learning algorithms in creating a smart recommendation system that can give personalized suggestions.

Through this project, one has gained knowledge of how Natural Language Processing works in terms of preprocessing, training, and recommendation.

Moreover, it has shown how Artificial Intelligence can be used in giving a better user experience.

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